

AMIR JAFARI, PhD

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RESEARCH PROFILE

Biosciences PhD with experience across molecular and cellular biology, behavioral pharmacology, and computational data analysis. Skilled in cell culture, flow-cytometry, PCR, animal handling, mouse behavioral phenotyping, and advanced Python-based analysis. My work connects molecular and cellular mechanisms with animal behavior, with current interests in time-series behavioral organization, causal inference, and predictive modeling in preclinical research.

CURRENT RESEARCH STATUS

Postdoctoral Fellow, Rio de Janeiro State University, Brazil

Apr 2025-present

- Research focus: Behavioral neuroscience with data-mining methods; *upcoming research paper is about Bupropion treatment*.

EDUCATION

PhD in Biosciences, Neurophysiology focus

Rio de Janeiro State University, Brazil, 2020-2024

- Dissertation title: Behavioral profiling of Swiss mice exposed to nicotine during adolescence.

MS.c in Medical Nanotechnology

Iran University of Medical Sciences, Iran, 2016-2019

- Thesis focus: Neural tissue engineering (maturation and morphology) with PC12 cell on developed scaffolds

BS.c in Chemistry

University of Tabriz, Iran, 2009-2013

CORE RESEARCH SKILLS

- Animal behavior: Open Field, Elevated Plus Maze, Conditioned Place Preference, Barnes maze, two-bottle choice, osmotic mini-pump implantation, drug injection (iv, im, sc, ip), animal husbandry.
- Computational analysis: Python, SPSS for statistical test (Classical and Modern)
- Biology lab: Cell culture (animal and human), IHC, QR-PCR, Primer design, ELISA, Western blotting, aseptic technique.
- Chemistry lab: Chemical synthesis, solution/buffer/reagent preparation, HPLC, FTIR, UV-Vis, TGA, basic material characterization.

SELECTED PUBLICATIONS

- **Jafari A. et al**, [Flexible Time-Series Analysis: A Dynamically Aware Method for Inferring Directed Dependencies in Behavioral Data. Behavioural Processes, 2026.](#)
- **Jafari A. et al**. Alignment Behavior of Nerve, Vascular, Muscle, Intestine Cells in 2D and 3D Strategies. WIREs Mechanisms of Disease, 2023.
- **Jafari A. et al**. Stimuli-responsive graphene-based nanomaterial through biological obstacles of tumor. FlatChem, 2022.
- **Jafari A. et al**. PC12 cells proliferation and morphological aspects: raffinose-grafted graphene oxide in silk fibroin-based scaffold. Materials Science and Engineering C, 2021.

For full list please [click here](#)

SELECTED HONORS & TRAINING

- PAPD-ranked Postdoctoral Fellow, Research Support Program, Rio De Janiro State University, *Brazil*, 2025.
- Nagoya University travel grant and full-stipend internship, Nagoya University, *Japan*, 2024.
- CAJAL Machine Learning for Neuroscience, Full stipend, Champalimaud Centre, *Portugal*, 2023.
- CAPES PhD scholarship, *Brazil*, 2020-2024.

References available upon request